KAIQIN KONG

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Education

University of California San Diego — La Jolla, CA
Master of Science in Computer Science — *Sep 2025 – Jun 2027*

Zhejiang University — Hangzhou, Zhejiang
Bachelor of Engineering in Industrial Design — *Sep 2020 – Jun 2024*

Open-Source Contributions

FastVideo Core Contributor

- Built end-to-end training recipes for world models, spanning bidirectional fine-tuning, diffusion-forcing and teacher-forcing SFT, and self-forcing DMD distillation.
- Built block-streaming inference for world models with stateful causal denoising, cached VAE decoding, and asynchronous multi-GPU execution.
- Added support for diverse world models and implemented new inference/training algorithms.

FastH3

- 4-step distilled MiniMax H3 (33B audio–video DiT) generating a **15-second 768p video with audio in 12.88 seconds** on 8× NVIDIA B200, up to 14× faster than base H3 on a single GPU.
- Integrated MiniMax H3 into FastVideo with T2VA, FL2VA, and Ref2VA pipelines, and cut inference memory and latency via FP8 loading, VAE memory reductions, and Triton kernel fusions.

FastWan-QAD

- Enabled end-to-end generation of a **5-second 480p video in 1.8 seconds** on a single NVIDIA GeForce RTX 5090, a 94× speedup over the original Wan2.1, and released three NVFP4/FP8 checkpoints.
- Co-designed the two-stage quantization-aware distillation recipe, combining Attn-QAT fine-tuning with 3-step DMD, and trained the released Wan2.1-T2V-1.3B checkpoints.

Dreamverse

- Open-source, self-hostable workspace for real-time "vibe directing" of 30-second 1080p scenes, generating each 5-second video with audio in ∼4.55 seconds on a single NVIDIA B200 GPU.
- Added end-to-end LoRA style controls across the backend API and React UI, and fixed LTX-2 audio–video desynchronization across multi-segment continuations.

Research

Matrix-Game 4.0: Engine-Grounded Neural Rendering for Real-Time Interactive Worlds

Jihai Zhang*, Runjia Qian*, **Kaiqin Kong***, Biao Jiang*, Puyi Wang, Siru Chen, Tianle Li, Guanjie Chen, Junyi Wu, Chubin Chen, Yuanhang Li, Kaichen Huang, Haobo Zhang, Wei Yu, Guibin Chen, Yu Cheng, and Yahui Zhou
Technical Report, 2026 *(* equal contribution)*

HeroWorld: Long-Horizon Action-Conditioned World Models for Third-Person Games

Mingjia Huo, **Kaiqin Kong**, Minshen Zhang, Shaoxiong Duan, Junda Su, Will Lin, and Hao Zhang
ICML 2026 F2S Workshop

Experience

Skywork AI Remote
Research Intern *Sep 2026 – Present*

- Research on game world models and digital humans, covering post-training and inference acceleration.

One Robot San Francisco, CA
Machine Learning Intern *Jun 2026 – Sep 2026*

- Trained and evaluated world models on robot trajectory data.

Hao AI Lab, UC San Diego La Jolla, CA
Research Intern *Nov 2025 – Present*

- Contribute to FastVideo, mainly on world models.

International Design Institute, Zhejiang University Hangzhou, Zhejiang
Research Intern *Jun 2024 – Jun 2025*

- Developed AI workflow powered by RAG over an HCI paper database to generate designs and evaluations.
- LLM fine-tuning and data processing pipeline development.

Zhejiang University Hangzhou, Zhejiang
Teaching Assistant | Computer Game Programming *Sep 2023 – Jan 2024*

- Guided students from initial proposal to final implementation of their games.
- Led weekly labs and graded assignments for students.